MBARARA UNIVERSITY OF SCIENCE AND TECHNOLOGY (MUST)

SWE2106: Internet Technologies and Web Design — Etivity 5

TEAM ACTIVITY REPORT, PROJECT PROPOSAL & SDLC EXECUTION PLAN

Project: Graduation Clearance Management System

PART 1: PROJECT PROPOSAL.

1. Problem Statement

At Mbarara University of Science and Technology (MUST), the traditional graduation clearance process is manual, slow, and paper-based. Graduating students must physically walk from one administrative office to another - including the Guild Office, Library, Finance, Head of Department (HOD), Faculty Dean, and Academic Registrar (AR) – to obtain physical signatures and stamps.

- **Long Delays & Queues:** Students spend days or weeks standing in long queues outside offices.
- **Lack of Visibility:** Students cannot track where their clearance file currently is or why it is stalled.
- **Loss of Records:** Physical clearance sheets can easily be misplaced, torn, or damaged.
- **Unmonitored Office Bottlenecks:** Faculty leadership lacks real-time oversight to detect which offices are delaying student clearance requests.

2. Target Users

- **Students:** Submit graduation clearance requests online, monitor real-time approval progress across departments, view specific rejection reasons, and resubmit directly to the rejecting office once issues are resolved.
- **Department Officers (Guild, Librarian, Finance, HOD, Dean, AR):** View pending student requests assigned strictly to their department stage, verify student records (dues, library books, departmental clearances), and approve or reject requests with mandatory feedback.
- **Faculty Head:** Oversees all departmental clearance activities across faculties, monitoring pending request counts and officer response logs to prevent bottlenecks and delays.

3. Project Objectives

- **Automate the Clearance Workflow:** Build a sequential digital workflow pipeline (Student > Guild -> Library -> Finance -> HOD -> Dean -> AR -> Completed).
- **Reduce Processing Time:** Eliminate physical movement and paper handling so students can complete clearance remotely.
- **Enable Targeted Resubmission Routing:** Ensure rejected requests return solely to the department that issued the rejection after the student resolves the issue, avoiding redundant re-approvals.
- **Enhance Accountability & Visibility:** Provide transparent status dashboards for students and real-time bottleneck analytics for the Faculty Head.

4. System Scope

In Scope: Authentication & Role-Based Access Control (JWT login for 8 distinct roles), automated state-machine workflow engine, mandatory rejection feedback, targeted resubmission routing, student notification inbox, and Faculty Head oversight analytics.

Out of Scope: Direct integration with third-party mobile money or banking payment gateways (outstanding tuition or library fines are settled offline at the bank prior to officer clearance).

5. Proposed Solution & Key Deliverables

A centralized full-stack web application built with Node.js and Express.js for the backend REST API, MySQL for relational database storage, and HTML5, CSS3, and Vanilla JavaScript for the responsive web interface.

Deliverable	**Target Timeline**
Project Proposal (Problem, Users, Objectives, Scope, Solution)	10th November 2026
Express Server & Core REST API Setup	12th November 2026
Documented Postman Collection (Testable Requests & Sample Data)	12th November 2026
GitHub Repository (Individual Member Branches & Pull Requests)	12th November 2026

Deliverable	Target Timeline
Live Database & Frontend Integration + Team Presentation	12th November 2026

PART II: TEAM MEETING REPORT.

Date & Time: Monday, 9th November 2026 | 10:00 AM – 1:30 PM EAT

Venue: MUST Computer Lab & Google Meet Hybrid Session

Attendance: All 10 Group Members (100% Attendance)

Agenda Item	Execution & Key Team Resolutions
1. Problem Identification & Selection	We evaluated registrar processes from the Etivity 5 brief and we selected Graduation Clearance due to the multi-office delays experienced by the graduating students at MUST.
2. Workflow & State-Machine Rules	We established the strict clearance order: Guild > Library > Finance > HOD > Dean > AR. We designed a targeted resubmission logic so rejected requests return only to the rejecting office, plus a Faculty Administrator oversight view.
3. Technology Stack & Tooling	We standardized on Node.js + Express (Backend), MySQL (Database), HTML5/CSS3/JS (Frontend), Figma (System Modeling & Wireframes), Postman (API Testing), and GitHub (Version Control).
4. Task Allocation & Git Workflow	We divided the project across 10 specialized roles. Mandated individual feature branches and reviewed Pull Requests on GitHub to verify every member's contribution.

Agenda Item	Execution & Key Team Resolutions
5. Research Sprints & Reconvene Plan	We structured alternating individual research breaks and group integration checkpoints leading up to the 10th Nov proposal and 12th Nov final demo.

PART III: SHARING OF ROLES AND TASKS.

Each team member was assigned a role and in some cases two members were working on the same thing, codebase responsibility, and GitHub feature branch to ensure clear accountability and parallel development:

MEMBER	Team Role	Module & Key Responsibilities	Relevant GitHub Files
GUMOSHABE MORRIS	**Team Leader & System Integrator**	Coordinates the development team, supervises implementation, manages integration of different modules and reviews code contributions before merging them into the main branch. Ensures that all system components work together.	`server.js`, `README.md`, Entire repository
AINEBYOONA BRAVE (with victor)	**Database Administrator**	Designed and implemented the MySQL database structure. Responsible for creating database tables, establishing relationships, defining primary and foreign keys, managing database connections and preparing sample records.	`database/`, `config/`
SEMYALO INNOCENT	**Authentication & Security Developer**	Implemented user authentication, login validation and access control. Responsible for verifying user credentials, managing authentication sessions or tokens, and ensuring that users can only access functions permitted by their assigned roles.	`authController.js`, `authMiddleware.js`, `login.html`

MEMBER	Team Role	Module & Key Responsibilities	Relevant GitHub Files
OTIM THOMAS ACQUINAS	**Backend & API Developer**	Developed the backend application using Node.js and Express.js. Responsible for configuring the server, defining API endpoints, handling HTTP requests and responses, and connecting the frontend to backend services.	`server.js`, `package.json`, `config/`
GUMOSHABE MORRIS	**Workflow Management Developer**	Implemented the graduation clearance workflow, ensuring that student requests move through the appropriate administrative stages. Responsible for handling approval, rejection, clearance status updates and resubmission processes.	`WorkFlow.js`, `js/`, `scripts/`
AHEREZA GODIUS	**Student Portal Developer**	Developed the student dashboard, enabling students to access their accounts, submit clearance requests, monitor clearance progress, view approval statuses and receive feedback on rejected requests.	`student-dashboard.html`, `js/`
OTIM THOMAS	**Departmental Officer Dashboard Developer**	Developed the officer dashboard used by departmental officers to access assigned clearance requests, search for students, view student information, verify clearance details and approve or reject requests with appropriate reasons.	`officer-dashboard.html`, `js/`
NKAMWESIGA A VICTOR	**Faculty Administrator Dashboard Developer**	Developed the Faculty Administrator dashboard to support institutional oversight of clearance activities. Responsible for displaying clearance statistics, monitoring departmental progress and identifying pending requests	`FacultyAdmin.html`, `js/`

MEMBER	Team Role	Module & Key Responsibilities	Relevant GitHub Files
		and possible administrative delays.	
AHUMUZA RACHEAL(with Innocent)	**Frontend & UI/UX Designing**	Designed and implemented the visual appearance of the system. Wireframes, Responsible for page layouts, navigation, responsive styling, forms, buttons, status indicators and consistent branding across the different dashboards.	`style.css`, `ham.png`, HTML files
AHEREZA GODIUS (with Innocent)	**Software Testing**	Conducted system testing to identify functional errors, validated user interactions and checked integration between frontend, backend and database components..	`scripts/`, `README.md`, `package.json`
NIMUSIIMA JANIPHER	**Documentation officer**	Compiled all agreed decisions and documented all implementations	`documentation`
KWAKA VIOLET SALLY	**Flow chart**	Drew the flow chart of the entire system	`violetsally17-patch-1` `png files`

PART IV: RESEARCH BREAKS AND RECONVENE INTEGRATION PLAN.

To balance deep technical learning with seamless system integration, the team operated in three focused sprints alternating between individual research and group reconvene reviews:

- Sprint 1 — Individual Research & Prototyping (Mon 2:00 PM – Tue 10:00 AM): Members conducted independent research on MySQL foreign key constraints and normalization (Member 2), Express middleware and JWT authentication (Members 1 & 3), Pencil wireframing and Workbench ERD modeling (Members 2 & 9), and Fetch API state updates (Members 4–8).
- Reconvene Session 1 — Architecture & Design Review (Tue 10:30 AM): Reviewed the database ERD and Pencil UI wireframes as a team. Key Feedback Integrated: Identified that forcing a rejected student to restart from the Guild Office wastes time;

added the last rejected by field to route resubmissions directly back to the rejecting department.

- Sprint 2 — Parallel Module Implementation (Tue 12:00 PM – Wed 4:00 PM): Members implemented their assigned Express routes, SQL queries, and HTML/JS dashboards independently on their respective GitHub branches.
- Reconvene Session 2 — Code Integration & Feedback (Wed 5:00 PM): Merged feature branches into main and tested cross-module workflows. Key Feedback Integrated: Added the Faculty Head oversight dashboard to monitor pending counts per office and prevent silent delays, and standardized institutional credentials (@role.must and Year/Course/Number/PS).
- Sprint 3 & Final Reconvene — QA Testing & Demo Rehearsal (Wed 7:00 PM – Thu 11:30 AM): Member 10 finalized the Postman test suite while all members verified edge cases and prepared technical explanations for their modules, followed by a full live presentation rehearsal on Thursday at 10:30 AM.

PART V: SOFTWARE DEVELOPMENT LIFE CYCLE (SDLC) PROJECT PLAN.

SDLC Phase	**Timeline**	**Activities, Tools & Outputs**
1. Planning & Requirements Analysis	9th – 10th Nov 2026	Analyzed manual clearance bottlenecks at MUST; defined functional requirements (sequential approvals, mandatory rejection comments, targeted resubmission) and non-functional requirements (JWT security, input validation). Output: Project Proposal.
2. Solution & System Design	10th Nov 2026	Designed the 7-table relational ERD and statemachine flowchart using Figma; sketched UI wireframes for Login, Student, Officer, and Faculty Head portals using Pencil; defined REST API endpoints and JSON contracts.

3. Code Implementation	10th – 11th Nov 2026	Developed MySQL schema and seed scripts; built Node.js/Express REST API with database transactions (BEGIN, COMMIT, ROLLBACK) for workflow integrity; implemented responsive HTML5, CSS3, and Vanilla JS dashboards connected via Fetch API.
4. Testing & Quality Assurance	11th – 12th Nov 2026	Executed unit, API, and integration tests using Postman (testing valid flows as well as invalid inputs like unauthorized stage approvals, empty rejection reasons, and duplicate requests) alongside browser User Acceptance Testing (UAT).
SDLC Phase	**Timeline**	**Activities, Tools & Outputs**
5. Deployment & Presentation	12th Nov 2026	Consolidated all 10 member Pull Requests into the main GitHub repository; prepared live local server, MySQL database, Postman collection, and frontend walkthrough for the class presentation.